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- Ivo Dell' Ambrogio, *The unitary symmetric monoidal model category of small C*-categories*
- Emily Riehl, *Homotopical categories: from model categories to* $(\infty, 1)$*-categories*

Theorem (Dell' Ambrogio)**.** C*Cat *is a model category.*

Definition. A *model category* is a bicomplete category with a *model structure*.

Definition (C*-category)**.** A C*-category is a category A which is a $\mathbb{C}$-linear category with conjugate-linear involutions $*: A(x,y) \to A(y,x)$ and complete norms $\|\cdot\| : A(x,y) \to [0,\infty)$ on each hom-set $A(x,y)$ such that $\|fg\| \le \|f\|\|g\|$ and $\|f^*f\| = \|f\|^2$ and $f^*f \ge 0$ are satisfied.

Example. A C*-algebra is a C*-category with a single object.

Example. The category Hilb defined such that objects are separable Hilbert spaces and morphisms are bounded linear operators is a C*-category.

Definition. The category C*Cat is the category of C*-categories whose morphisms are given by $*$-functors.

Definition. Two C*-categories A and B are called *unitary equivalent* if there are $*$-functors $F : A \to B$ and $G : B \to A$ such that $GF \simeq \mathrm{id}_A$ and $FG \simeq \mathrm{id}_B$. Here, $GF \simeq \mathrm{id}_A$ means that for each morphism $f : a \to a'$ in A, we have unitaries $u_a : GF(a) \to a$ and $u_{a'} : GF(a') \to a'$ such that $fu_a = u_{a'}GF(f)$. The unitarity of u_a means that $u_a^*u_a = \mathrm{id}_{GF(a)}$ and $u_au_a^* = \mathrm{id}_a$.

Definition. We define a model structure on C*Cat as follows:

- weak equivalences are unitary equivalences,
- cofibrations are $*$-functors injective on objects,
- fibrations are $*$-functors such that whenever there is a unitary (equivalence) $u : b \sim Fa'$ for some $a' \in A$ we have $a \in A$ and unitary $v : a \to a'$ such that $Fv = u$, and hence $Fa = b$.

Proof of Theorem. **Axiom 1: Two-out-of-Three law**

Clear.

Axiom 2: Retracts

Consider the diagram in C*Cat

$$\begin{array}{ccccc} A & \xrightarrow{P} & A' & \xrightarrow{Q} & A \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle f} \\ B & \xrightarrow{I} & B' & \xrightarrow{J} & B \end{array}$$

such that each row is composed to the indentites.

Assuming g is a fibration, we want to show f is a fibration. Let $u : b \sim fa'$ for $b \in B$ and $a' \in A$. Then, $Ib \sim Ifa' = gPa'$. Since g is a fibration, there is $\widetilde{a} \in A'$ with a unitary $\widetilde{a} \to Pa'$ and $g\widetilde{a} = Ib$. Then, $Q\widetilde{a} \sim QPa' = a'$ and $fQ\widetilde{a} = Jg\widetilde{a} = JIb = b$.

Cofibrations are clear.

For weak equivalences, if h is the 'weak inverse' of g and $\widetilde{f} := QhI$, then f and $\widetilde{f}$ show that f is unitary.

Axiom 3: Lifting

Consider

$$\begin{array}{ccc} A & \xrightarrow{P} & B \\ \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} \\ C & \xrightarrow{Q} & D \end{array}$$

Suppose first α is an acylic cofibration and β is a fibration. Take a weak inverse $\widetilde{\alpha} : C \to A$ of α. Since for $c \in C$ we have $Q(c) \sim Q\alpha\widetilde{\alpha}(c) = \beta P\widetilde{\alpha}(c) \in D$, there is $b \in B$ and unitary $u_b : b \to P\widetilde{\alpha}(c)$ and $\beta(b) = Q(c)$. Define $\gamma : C$ as follows. Let $\gamma(c) := b$. For $f : c \to c'$ in C, we define $\gamma(f) : b \to b'$ such that

$$b \sim P\widetilde{\alpha}(c) \xrightarrow{P\widetilde{\alpha}(f)} P\widetilde{\alpha}(c') \sim b'.$$

Now we check the commutation. Since α is an acylic cofibration, by Lemma in the paper, we may assume $\widetilde{\alpha}\alpha = \mathrm{id}_A$. Then, $\beta P(a) = \beta P\widetilde{\alpha}\alpha(a) = \beta P$

$\square$